You said:

calculus homework 35

ChatGPT said:

Let's work through **Jerry's AP Calculus AB Homework #35** using *James Stewart, Calculus, 8th Edition*.

Please send the **assigned page numbers and problem numbers**, or upload a photo of the homework.

For each problem, I'll provide the problem statement, a step-by-step solution, the key calculus concepts and common mistakes to watch for.

You said:

problem 24 statement and solution

ChatGPT said:

Could you provide the **page number or section** for Problem 24 in Homework #35?

I know Jerry uses *James Stewart, Calculus, 8th Edition*. I just need the page number (or a photo) to identify the correct problem and give its statement, step-by-step solution, and key concepts.

You said:

p193 #24-30 even, 31-33 & p198-199 #76-81 also do #7-9 from RR worksheet
Hint for 77: differentiate two formulas Area & Volume

ChatGPT said:

Here is Jerry's **AP Calculus Homework #35** assignment:

Pages	Problems
193	24, 26, 28, 30, 31, 32, 33
198–199	76, 77, 78, 79, 80, 81

Related Rates (RR) worksheet	7, 8, 9

Teacher's hint for #77: Differentiate both the area and volume formulas.

I can work through each problem with its statement, step-by-step solution, and key concepts.

For **Problem 24 on page 193**, please upload a photo of the page. I couldn't reliably match that page to the correct edition of Jerry's Stewart textbook, and I don't want to solve the wrong problem.

You said:

problem 24 statement and solution page 193

ChatGPT said:

Homework #35 — Page 193, Problem 24

Textbook: James Stewart, *Calculus*, 8th Edition
Section 2.9: Linear Approximations and Differentials

Problem statement

Use a linear approximation to estimate:

$$\boxed{\frac{1}{4.002}}$$

Step-by-step solution

Step 1: Choose a function and a convenient value.

Rewrite the expression:

$$\frac{1}{4.002} = \frac{1}{2(2.001)}$$

Choose:

$$f(x) = \frac{1}{2x}, \qquad a = 2$$

Here, $x = 2.001$, so:

$$\Delta x = 2.001 - 2 = 0.001$$

Step 2: Find the derivative.

$$f(x) = \frac{1}{2}x^{-1}$$

Using the power rule:

$$f'(x) = -\frac{1}{2}x^{-2} = -\frac{1}{2x^2}$$

At $x = 2$:

$$f'(2) = -\frac{1}{2(2)^2} = -\frac{1}{8}$$

Step 3: Apply the linear approximation formula.

Recall:

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

We know:

$$f(2) = \frac{1}{4}, \qquad f'(2) = -\frac{1}{8}$$

Therefore:

$$L(2.001) = \frac{1}{4} - \frac{1}{8}(0.001)$$

$$= 0.25 - 0.000125$$

$$\boxed{L(2.001) = 0.249875}$$

Final answer

$$\boxed{\frac{1}{4.002} \approx 0.249875}$$

This matches the published solution. GradeSaver

Key concept for Jerry

Linear approximation uses the tangent line at a convenient point to estimate a nearby function value:

$$\boxed{f(a + \Delta x) \approx f(a) + f'(a)\Delta x}$$

Important: The approximation becomes particularly accurate when Δx is small. Here, $\Delta x = 0.001$, so the estimate is very close to the actual value.

Common mistake: Using $x = 4$ with $f(x) = 1/(2x)$. The function and the chosen input must match the original expression.

You said:

jerry use f(x) = 1/4x, x =1 , delta x = 0.002

ChatGPT said:

Jerry's approach is creative. **Choosing $f(x) = \frac{1}{4x}$ with $a = 1$ is valid**, but his $\Delta x = 0.002$ needs correction.

We want to approximate:

$$\frac{1}{4.002}$$

Step 1: Check Jerry's setup

Jerry chose:

$$f(x) = \frac{1}{4x}, \qquad a = 1$$

To reproduce the original expression, we need:

$$4x = 4.002$$

$$x = 1.0005$$

Therefore:

$$\boxed{\Delta x = 1.0005 - 1 = 0.0005}$$

Not 0.002.

Step 2: Differentiate

$$f'(x) = -\frac{1}{4x^2}$$

At $a = 1$:

$$f(1) = \frac{1}{4}, \qquad f'(1) = -\frac{1}{4}$$

Step 3: Linear approximation

$$f(a + \Delta x) \approx f(a) + f'(a)\Delta x$$

$$\approx \frac{1}{4} - \frac{1}{4}(0.0005)$$

$$\boxed{0.249875}$$

My assessment of Jerry's method

- **Function selection:** Correct.
- **Base point $a = 1$:** Correct and convenient.
- **$\Delta x = 0.002$:** Incorrect; it should be 0.0005.
- **Overall reasoning:** Good, provided he understands how the original denominator maps to the input x.

The important habit for Jerry is to **substitute his chosen x back into $f(x)$** before calculating the derivative.

For example, his original $x = 1.002$ gives:

$$f(1.002) = \frac{1}{4(1.002)} = \frac{1}{4.008}$$

which is not the quantity the problem asks him to approximate.

This is exactly the kind of small setup error that careful checking can catch before it costs points on an AP Calculus test.

You said:

problem 26 statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 26

Textbook: James Stewart, *Calculus*, 8th Edition
Section 2.9: Linear Approximations and Differentials

Problem statement

Use a linear approximation (or differentials) to estimate:

$$\boxed{\sqrt{100.5}}$$

Step-by-step solution

Step 1: Choose a function and a convenient point.

$$f(x) = \sqrt{x} = x^{1/2}$$

Choose $a = 100$, because $\sqrt{100} = 10$.

$$x = 100.5, \qquad \Delta x = 0.5$$

Step 2: Find the derivative.

Using the power rule:

$$f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

At $x = 100$:

$$f'(100) = \frac{1}{2\sqrt{100}} = \frac{1}{20}$$

Step 3: Apply linear approximation.

Recall:

$$\boxed{f(a + \Delta x) \approx f(a) + f'(a)\Delta x}$$

Substitute:

$$\sqrt{100.5} \approx 10 + \frac{1}{20}(0.5)$$

$$= 10 + 0.025$$

Final answer

$$\boxed{\sqrt{100.5} \approx 10.025}$$

This matches the textbook solution. Test Bank Fan

Key concept for Jerry

The trick is choosing a nearby number whose square root is easy to calculate.

- 100.5 is close to 100.
- $\sqrt{100} = 10$.
- The derivative tells us how much the square root changes when x increases by 0.5.

Quick check: Since $100.5 > 100$, the answer should be slightly greater than 10.

The same problem can also be solved using differentials:

$$dy = f'(x)\,dx$$

$$dy = \frac{1}{2\sqrt{100}}(0.5) = 0.025$$

Therefore:

$$\boxed{y + dy = 10.025}$$

You said:

problem 28 and statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 28

Problem statement

Use a linear approximation (or differentials) to estimate:

$$\boxed{\cos 29^\circ}$$

Step 1: Choose a function and a convenient point

Let

$$f(x) = \cos x$$

Choose $a = 30^\circ = \frac{\pi}{6}$, because we know:

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

We want $x = 29^\circ$, so:

$$\Delta x = 29^\circ - 30^\circ = -1^\circ$$

Important: Convert degrees to radians before using derivatives.

$$\boxed{\Delta x = -\frac{\pi}{180}}$$

Step 2: Find the derivative

$$f'(x) = -\sin x$$

At $a = \frac{\pi}{6}$:

$$f'(a) = -\sin\frac{\pi}{6} = -\frac{1}{2}$$

Step 3: Apply linear approximation

Recall:

$$\boxed{f(a+\Delta x) \approx f(a) + f'(a)\Delta x}$$

Substitute:

$$\cos 29^\circ \approx \frac{\sqrt{3}}{2} + \left(-\frac{1}{2}\right)\left(-\frac{\pi}{180}\right)$$

Simplify:

$$= \frac{\sqrt{3}}{2} + \frac{\pi}{360}$$

$$\approx 0.866025 + 0.008727$$

Final answer

$$\boxed{\cos 29^\circ \approx 0.874752}$$

Rounded to three decimal places:

$$\boxed{0.875}$$

This agrees with the textbook solution. GradeSaver

Key concepts for Jerry

1. **Choose a nearby familiar angle:** 30° is convenient because its sine and cosine are known exactly.
2. **Convert degrees to radians:** Differentiation formulas such as $\frac{d}{dx}\cos x = -\sin x$ assume radians.
3. **Check the direction:** Since cosine decreases between 0° and 90°, $\cos 29^\circ$ must be slightly greater than $\cos 30^\circ$.

Common AP Calculus mistake: Using $\Delta x = -1$ instead of $-\pi/180$. That produces a dramatically incorrect estimate.

You said:

problem 30 statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 30

Problem statement

Explain, using linear approximations or differentials, why the following approximation is reasonable:

$$\boxed{\sqrt{4.02} \approx 2.005}$$

Step 1: Choose a function and a convenient point

Let:

$$f(x) = \sqrt{x} = x^{1/2}$$

Choose $a = 4$, because $\sqrt{4} = 2$.

$$x = 4.02, \qquad \Delta x = 4.02 - 4 = 0.02$$

Step 2: Find the derivative

$$f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

At $a = 4$:

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

Step 3: Apply linear approximation

Recall:

$$\boxed{f(a + \Delta x) \approx f(a) + f'(a)\Delta x}$$

Substitute:

$$\sqrt{4.02} \approx \sqrt{4} + \frac{1}{4}(0.02)$$

$$= 2 + 0.005$$

$$\boxed{\sqrt{4.02} \approx 2.005}$$

Final answer

The approximation is reasonable because 4.02 is very close to 4, so the tangent line at $x = 4$ provides an accurate estimate of $\sqrt{4.02}$.

Key concept for Jerry

Notice the difference between Problems 28 and 30:

- **Problem 28:** Find an approximation using linearization.
- **Problem 30:** Explain why an approximation is reasonable using linearization.

In both cases, the mathematics is the same:

$$\boxed{\Delta y \approx dy = f'(a)\Delta x}$$

AP Calculus tip: For explanation questions, Jerry should show the derivative calculation and explicitly connect the result to the given approximation—not just state that the numbers are close.

You said:

problem 31 statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 31

Problem statement (paraphrased)

A cube has an edge length measured as **30 cm**, with a possible measurement error of **0.1 cm**.

(a) Use differentials to estimate the maximum possible error, relative error, and percentage error in calculating the cube's volume.

(b) Find the same three errors for its surface area. Test Bank Fan

Part (a): Volume

Step 1: Write the volume formula.

$$V = x^3$$

Given:

$$x = 30 \text{ cm}, \qquad |dx| = 0.1 \text{ cm}$$

Step 2: Differentiate.

$$\boxed{dV = 3x^2\,dx}$$

Step 3: Calculate maximum possible error.

$$|dV| = 3(30)^2(0.1)$$

$$\boxed{|dV| = 270 \text{ cm}^3}$$

Step 4: Calculate relative error.

$$\text{Relative error} \approx \frac{|dV|}{V}$$

Since $V = 30^3 = 27000$:

$$\frac{270}{27000} = \boxed{0.01}$$

Step 5: Calculate percentage error.

$$0.01 \times 100\% = \boxed{1\%}$$

Part (b): Surface area

Step 1: Write the surface area formula.

$$S = 6x^2$$

Step 2: Differentiate.

$$\boxed{dS = 12x\,dx}$$

Step 3: Calculate maximum possible error.

$$|dS| = 12(30)(0.1)$$

$$\boxed{|dS| = 36\ \text{cm}^2}$$

Step 4: Calculate relative error.

$$S = 6(30)^2 = 5400$$

$$\frac{|dS|}{S} = \frac{36}{5400} \approx \boxed{0.00667}$$

Step 5: Calculate percentage error.

$$0.00667 \times 100\% \approx \boxed{0.667\%}$$

Final answers

Error	(a) Volume	(b) Surface area

Maximum possible error	270 cm³	36 cm²
Relative error	0.01	0.00667
Percentage error	1%	0.667%

Key concepts for Jerry

The important difference is between **absolute error** and **relative error**.

$$\text{Absolute error} \approx |dy|$$

$$\text{Relative error} \approx \frac{|dy|}{|y|}$$

$$\text{Percentage error} \approx \frac{|dy|}{|y|} \times 100\%$$

There is also a useful shortcut:

For $y = Cx^n$,

$$\boxed{\frac{dy}{y} = n\frac{dx}{x}}$$

That means the relative error is multiplied by the exponent.

- Volume $V = x^3$: relative error is approximately **3 times** the relative error in edge length.
- Surface area $S = 6x^2$: relative error is approximately **2 times** the relative error in edge length.

This is why volume has a greater percentage error than surface area for the same edge-length measurement error.

You said:

problem 32 statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 32

Problem statement (paraphrased)

The radius of a circular disk is measured as **24 cm**, with a maximum measurement error of **0.2 cm**.

(a) Use differentials to estimate the maximum possible error in the calculated area.

(b) Find the relative error and percentage error. Vaia

Part (a): Maximum possible error in area

Step 1: Write the area formula.

$$A = \pi r^2$$

Given:

$$r = 24 \text{ cm}, \qquad |dr| = 0.2 \text{ cm}$$

Step 2: Differentiate.

$$\frac{dA}{dr} = 2\pi r$$

Therefore:

$$\boxed{dA = 2\pi r \, dr}$$

Step 3: Substitute the values.

$$|dA| = 2\pi(24)(0.2)$$

$$= 9.6\pi$$

$$\boxed{|dA| \approx 30.16 \text{ cm}^2}$$

Part (b): Relative and percentage error

Step 1: Calculate the area.

$$A = \pi(24)^2 = 576\pi$$

Step 2: Calculate relative error.

$$\text{Relative error} \approx \frac{|dA|}{A}$$

$$= \frac{9.6\pi}{576\pi}$$

$$\boxed{\frac{1}{60} \approx 0.01667}$$

Step 3: Calculate percentage error.

$$0.01667 \times 100\%$$

$$\boxed{1.67\%}$$

Final answers

Quantity	Answer

(a) Maximum error in area	$9.6\pi \approx 30.16\ \text{cm}^2$
(b) Relative error	0.01667
(b) Percentage error	1.67%

Key concept for Jerry

A useful shortcut comes directly from differentiation:

$$A = \pi r^2$$

$$dA = 2\pi r\, dr$$

Divide the two equations:

$$\boxed{\frac{dA}{A} = 2\frac{dr}{r}}$$

Interpretation: The percentage error in a circle's area is approximately twice the percentage error in its radius.

Here:

$$\frac{0.2}{24} \times 100\% = 0.8333\%$$

Double it:

$$2(0.8333\%) = \boxed{1.67\%}$$

This is the same principle Jerry used in Problem 31 for the cube.

You said:

problem 33 statement and solution

ChatGPT said:

Homework #35 — Page 193, Problem 33

Textbook: James Stewart, *Calculus*, 8th Edition
Section 2.9: Linear Approximations and Differentials

Problem statement (paraphrased)

The circumference of a sphere was measured to be **84 cm**, with a possible measurement error of **0.5 cm**.

(a) Use differentials to estimate the maximum error in the calculated surface area. What is the relative error?

(b) Use differentials to estimate the maximum error in the calculated volume. What is the relative error? Test Banks

Given

$$C = 84 \text{ cm}, \qquad |dC| = 0.5 \text{ cm}$$

We know:

$$C = 2\pi r \implies r = \frac{C}{2\pi}$$

Part (a): Surface area

Step 1: Express surface area in terms of circumference.

$$S = 4\pi r^2$$

Substitute $r = \frac{C}{2\pi}$:

$$S = 4\pi \left(\frac{C}{2\pi}\right)^2$$

Simplify:

$$\boxed{S = \frac{C^2}{\pi}}$$

Step 2: Differentiate.

$$\boxed{dS = \frac{2C}{\pi}\,dC}$$

Step 3: Estimate the maximum error.

$$|dS| = \frac{2(84)}{\pi}(0.5)$$

$$\boxed{|dS| = \frac{84}{\pi} \approx 26.74 \text{ cm}^2}$$

Step 4: Calculate relative error.

$$\frac{|dS|}{S} = \frac{2|dC|}{C}$$

$$= \frac{2(0.5)}{84}$$

$$\boxed{\text{Relative error} = \frac{1}{84} \approx 0.0119}$$

Percentage error:

$$\boxed{1.19\%}$$

Part (b): Volume

Step 1: Express volume in terms of circumference.

$$V = \frac{4}{3}\pi r^3$$

Substitute $r = \frac{C}{2\pi}$:

$$V = \frac{4}{3}\pi \left(\frac{C}{2\pi}\right)^3$$

Simplify:

$$\boxed{V = \frac{C^3}{6\pi^2}}$$

Step 2: Differentiate.

$$\boxed{dV = \frac{C^2}{2\pi^2}\, dC}$$

Step 3: Estimate the maximum error.

$$|dV| = \frac{84^2}{2\pi^2}(0.5)$$

$$= \frac{1764}{\pi^2}$$

$$\boxed{|dV| \approx 178.73 \text{ cm}^3}$$

Step 4: Calculate relative error.

$$\frac{|dV|}{V} = \frac{3|dC|}{C}$$

$$= \frac{3(0.5)}{84}$$

$$\boxed{\text{Relative error} = \frac{1}{56} \approx 0.01786}$$

Percentage error:

$$\boxed{1.79\%}$$

Final answers

Quantity	(a) Surface area	(b) Volume
Maximum error	26.74 cm^2	178.73 cm^3
Relative error	0.0119	0.01786
Percentage error	1.19%	1.79%

Key concept for Jerry

This is similar to Problems 31 and 32, except that **the measured quantity is circumference, not radius**.

An efficient strategy is to express surface area and volume directly in terms of circumference before differentiating.

The general shortcut is:

$$\boxed{y = kx^n \implies \frac{dy}{y} = n\frac{dx}{x}}$$

Here, surface area is proportional to C^2, while volume is proportional to C^3.

Therefore:

$$\boxed{\frac{dS}{S} = 2\frac{dC}{C}} \qquad \boxed{\frac{dV}{V} = 3\frac{dC}{C}}$$

Important test-taking insight: The volume's percentage error is greater because volume depends on the third power of the measured circumference, whereas surface area depends on its square.

You said:

problem 76 statement and solution

ChatGPT said:

Homework #35 — Page 198, Problem 76

Textbook: James Stewart, *Calculus*, 8th Edition
Chapter 2 Review — Derivatives

Problem statement (paraphrased)

The cost of producing x items is modeled by:

$$\boxed{C(x) = 920 + 2x - 0.02x^2 + 0.00007x^3}$$

The problem asks you to:

(a) Find the marginal cost function.

(b) Find the marginal cost at a production level of 100 items and interpret it.

(c) Find the actual cost of producing the 101st item and compare it with the marginal cost estimate. Calculus (8th ...

Part (a): Find the marginal cost function

Marginal cost is the derivative of total cost:

$$MC(x) = C'(x)$$

Differentiate each term:

$$C'(x) = 0 + 2 - 0.04x + 0.00021x^2$$

Therefore:

$$\boxed{C'(x) = 2 - 0.04x + 0.00021x^2}$$

Part (b): Marginal cost at 100 items

Substitute $x = 100$:

$$C'(100) = 2 - 0.04(100) + 0.00021(100)^2$$

$$= 2 - 4 + 2.1$$

$$\boxed{C'(100) = 0.10}$$

Interpretation: When 100 items have been produced, the cost of producing one additional item is approximately \$0.10.

Part (c): Actual cost of the 101st item

The actual additional cost is:

$$\boxed{C(101) - C(100)}$$

Rather than evaluating two large expressions separately, subtract algebraically:

$$\begin{aligned} C(101) - C(100) = {} & 2(101 - 100) \\ & - 0.02(101^2 - 100^2) \\ & + 0.00007(101^3 - 100^3) \end{aligned}$$

Use:

$$101^2 - 100^2 = 201$$

$$101^3 - 100^3 = 30301$$

Therefore:

$$= 2 - 0.02(201) + 0.00007(30301)$$

$$= 2 - 4.02 + 2.12107$$

$$\boxed{C(101) - C(100) = \$0.10107}$$

Final answers

Part	Answer
(a) Marginal cost	$C'(x) = 2 - 0.04x + 0.00021x^2$
(b) At 100 items	\$0.10 per item
(c) Actual cost of 101st item	\$0.10107

Key concept for Jerry

This problem connects **derivatives to linear approximation**, the topic of his previous homework.

Recall:

$$\boxed{\Delta C \approx C'(x)\Delta x}$$

When production increases from 100 to 101:

$$\Delta x = 1$$

So:

$$\Delta C \approx C'(100)(1) = \$0.10$$

Compare:

- Marginal cost estimate: \$0.10000
- Actual additional cost: \$0.10107

The difference is only \$0.00107.

Important AP Calculus concept: The derivative $C'(100)$ is an instantaneous rate of change, not exactly the cost of the 101st item. It provides an approximation to that additional cost.

You said:

problem 77 statement and solution

ChatGPT said:

Homework #35 — Page 199, Problem 77

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Volume and Surface Area

Problem statement (paraphrased)

The volume of a cube is increasing at a rate of **10 cm³/min**. How fast is its surface area increasing when the length of an edge is **30 cm**? GradeSaver

Given

Let x be the edge length, V the volume, and S the surface area.

$$\frac{dV}{dt} = 10 \text{ cm}^3/\text{min}$$

$$x = 30 \text{ cm}$$

Find:

$$\boxed{\frac{dS}{dt} = ?}$$

Step 1: Write the volume and surface area formulas

For a cube:

$$\boxed{V = x^3}$$

$$\boxed{S = 6x^2}$$

As Jerry's teacher hinted, we need to differentiate **both formulas**.

Step 2: Differentiate the volume formula

Starting with:

$$V = x^3$$

Differentiate both sides with respect to time t, using the Chain Rule:

$$\boxed{\frac{dV}{dt} = 3x^2 \frac{dx}{dt}}$$

Substitute the given volume rate:

$$10 = 3x^2 \frac{dx}{dt}$$

Solve for the rate at which the edge length increases:

$$\frac{dx}{dt} = \frac{10}{3x^2}$$

At $x = 30$:

$$\frac{dx}{dt} = \frac{10}{3(30)^2}$$

$$\boxed{\frac{dx}{dt} = \frac{1}{270} \text{ cm/min}}$$

Step 3: Differentiate the surface area formula

Starting with:

$$S = 6x^2$$

Differentiate with respect to time:

$$\boxed{\frac{dS}{dt} = 12x \frac{dx}{dt}}$$

Substitute $x = 30$ and $\frac{dx}{dt} = \frac{1}{270}$:

$$\frac{dS}{dt} = 12(30)\left(\frac{1}{270}\right)$$

$$= \frac{360}{270}$$

$$\boxed{\frac{dS}{dt} = \frac{4}{3}\ \text{cm}^2/\text{min}}$$

Final answer

$$\boxed{\frac{dS}{dt} = \frac{4}{3}\ \text{cm}^2/\text{min}}$$

The cube's surface area is increasing at approximately **1.33 cm² per minute**.

Key concept for Jerry

This problem involves three related rates:

Given: Volume rate

$$\frac{dV}{dt} = 10$$

↓ Differentiate $V = x^3$

Find: Edge-length rate

$$\frac{dx}{dt} = \frac{1}{270}$$

↓ Differentiate $S = 6x^2$

Answer: Surface-area rate

$$\frac{dS}{dt} = \frac{4}{3}$$

Important AP Calculus lesson: When two quantities are related through a third changing quantity, use the Chain Rule to connect their rates.

Jerry can also solve it faster by eliminating $\frac{dx}{dt}$:

$$\frac{dS}{dt} = 12x\left(\frac{1}{3x^2}\frac{dV}{dt}\right)$$

$$\boxed{\frac{dS}{dt} = \frac{4}{x}\frac{dV}{dt}}$$

At $x = 30$:

$$\frac{dS}{dt} = \frac{4}{30}(10) = \boxed{\frac{4}{3}}$$

This shortcut is useful for checking his work.

You said:

problem 78 statement and solution

ChatGPT said:

Homework #35 — Page 199, Problem 78

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Water Rising in a Cone

Problem statement (paraphrased)

A paper cup is shaped like a cone with a height of **10 cm** and a radius of **3 cm** at the top.

Water is poured into the cup at a rate of **2 cm³/s**.

How fast is the water level rising when the water is 5 cm deep? College Math T...

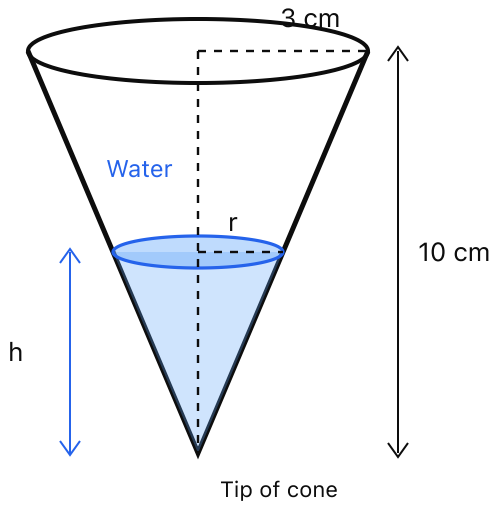


Diagram of the cup and the rising water level (not to scale).

Step 1: Identify the given values

Let:

- V = volume of water (cm³)
- h = water depth (cm)
- r = radius of the water surface (cm)

We know:

$$\frac{dV}{dt} = 2 \text{ cm}^3/\text{s}$$

When:

$$h = 5 \text{ cm}$$

Find:

$$\boxed{\frac{dh}{dt} = ?}$$

Step 2: Write the volume formula

For a cone:

$$\boxed{V = \frac{1}{3}\pi r^2 h}$$

Important: Both r and h change as water enters the cup.

We should express r in terms of h before differentiating.

Step 3: Use similar triangles

The full cone has radius 3 cm and height 10 cm.

The water forms a smaller, similar cone.

Therefore:

$$\frac{r}{h} = \frac{3}{10}$$

Solve for r:

$$\boxed{r = \frac{3}{10}h}$$

This is the key geometric relationship.

Step 4: Substitute into the volume formula

Start with:

$$V = \frac{1}{3}\pi r^2 h$$

Substitute $r = \frac{3}{10}h$:

$$V = \frac{1}{3}\pi \left(\frac{3}{10}h\right)^2 h$$

$$= \frac{1}{3}\pi \left(\frac{9}{100}h^2\right) h$$

Simplify:

$$\boxed{V = \frac{3\pi}{100}h^3}$$

Now volume depends only on h.

Step 5: Differentiate with respect to time

$$V = \frac{3\pi}{100}h^3$$

Using the Chain Rule:

$$\frac{dV}{dt} = \frac{3\pi}{100}(3h^2)\frac{dh}{dt}$$

$$\boxed{\frac{dV}{dt} = \frac{9\pi}{100}h^2\frac{dh}{dt}}$$

Step 6: Substitute the known values

Given:

$$\frac{dV}{dt} = 2, \qquad h = 5$$

$$2 = \frac{9\pi}{100}(5)^2\frac{dh}{dt}$$

Simplify:

$$2 = \frac{225\pi}{100}\frac{dh}{dt}$$

$$2 = \frac{9\pi}{4}\frac{dh}{dt}$$

Solve:

$$\frac{dh}{dt} = \frac{2(4)}{9\pi}$$

$$\boxed{\frac{dh}{dt} = \frac{8}{9\pi}}$$

Final answer

$$\boxed{\frac{dh}{dt} = \frac{8}{9\pi} \approx 0.283 \text{ cm/s}}$$

The water level is rising at approximately **0.283 centimeters per second** when it is 5 cm deep.

Key concept for Jerry

The hardest part of this problem is usually **not differentiation, but eliminating the extra variable r**.

Compare Problems 77 and 78:

Problem 77: Cube	Problem 78: Cone
$V = x^3$	$V = \frac{1}{3}\pi r^2 h$
One changing dimension x	Two changing dimensions r, h
Differentiate directly	Use similar triangles first
Chain Rule	Geometry + Chain Rule

Common mistake: Substituting $h = 5$ before differentiating. Jerry should keep h as a variable until after finding $\frac{dV}{dt}$.

Physical check: As the cone gets wider toward the top, the same amount of added water causes a smaller increase in water height. Therefore, $\frac{dh}{dt}$ decreases as h increases.

You said:

problem 79 statement and solution

ChatGPT said:

Homework #35 — Page 199, Problem 79

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Pythagorean Theorem

Problem statement (paraphrased)

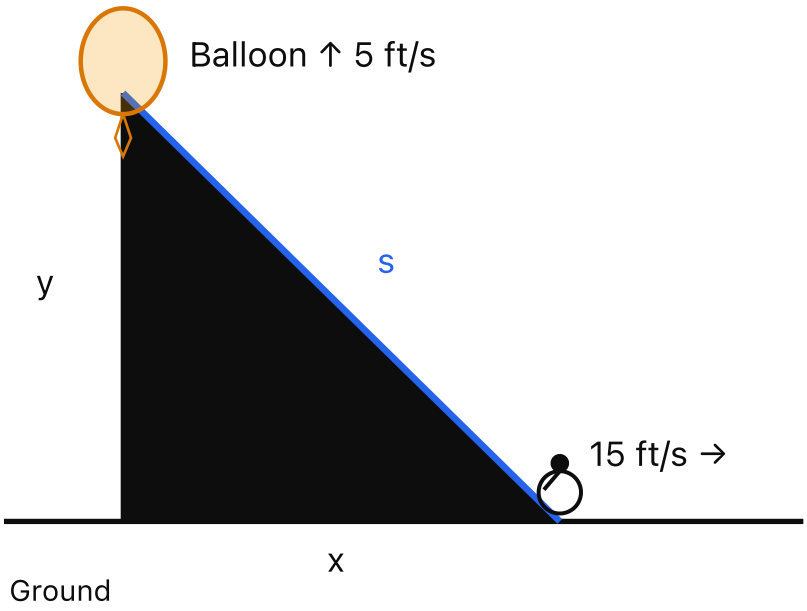
A balloon rises vertically at a constant speed of **5 ft/s**. A boy rides his bicycle along a straight road at **15 ft/s**.

When the boy passes directly beneath the balloon, the balloon is 45 ft above him.

How fast is the distance between the boy and the balloon increasing 3 seconds later?

College Math T...

Step 1: Draw a diagram



Let:

- x = horizontal distance traveled by the boy
- y = height of the balloon above the road
- s = distance between the boy and balloon

Step 2: Identify the given rates

$$\frac{dx}{dt} = 15 \text{ ft/s}$$

$$\frac{dy}{dt} = 5 \text{ ft/s}$$

At $t = 0$:

$$x = 0, \qquad y = 45$$

We want:

$$\boxed{\frac{ds}{dt} \text{ when } t = 3}$$

Step 3: Find the distances after 3 seconds

The boy travels horizontally:

$$x = 15(3) = 45 \text{ ft}$$

The balloon rises another $5(3) = 15$ ft:

$$y = 45 + 5(3) = 60 \text{ ft}$$

So at $t = 3$:

$$\boxed{x = 45, \qquad y = 60}$$

Step 4: Use the Pythagorean Theorem

The three distances form a right triangle:

$$\boxed{s^2 = x^2 + y^2}$$

Substitute $x = 45$ and $y = 60$:

$$s^2 = 45^2 + 60^2$$

$$s^2 = 2025 + 3600 = 5625$$

$$\boxed{s = 75 \text{ ft}}$$

Step 5: Differentiate with respect to time

Start with:

$$s^2 = x^2 + y^2$$

Using the Chain Rule:

$$2s\frac{ds}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

Divide by $2s$:

$$\boxed{\frac{ds}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{s}}$$

Step 6: Substitute the values

$$\frac{ds}{dt} = \frac{45(15) + 60(5)}{75}$$

$$= \frac{675 + 300}{75}$$

$$= \frac{975}{75}$$

Final answer

$$\boxed{\frac{ds}{dt} = 13 \text{ ft/s}}$$

Three seconds after the boy passes beneath the balloon, the distance between them is **increasing at 13 ft/s**.

Key concepts for Jerry

This is a classic AP Calculus related-rates problem.

1. Distinguish distance from rate.

At $t = 3$:

$$x = 45, \quad y = 60, \quad s = 75$$

But the rates are:

$$\frac{dx}{dt} = 15, \quad \frac{dy}{dt} = 5, \quad \frac{ds}{dt} = 13$$

2. Remember the Chain Rule.

$$\frac{d}{dt}(x^2) = 2x\frac{dx}{dt}$$

Not just $2x$, because x changes with time.

3. Do not simply add the speeds.

It would be incorrect to write:

$$\frac{ds}{dt} = 15 + 5 = 20$$

The motions are perpendicular, so we must account for the geometry.

4. Useful shortcut for checking:

Since $x = 15t$ and $y = 45 + 5t$, we can write:

$$s(t) = \sqrt{(15t)^2 + (45 + 5t)^2}$$

Differentiate this function directly to obtain the same result.

My suggestion: Jerry should be able to derive the formula from $s^2 = x^2 + y^2$ rather than memorizing it. This strategy also works for ladder, moving-car, and airplane related-rates problems.

You said:

problem 80 statement and solution

ChatGPT said:

Homework #35 — Page 199, Problem 80

Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Chain Rule and Related Rates

Problem statement (paraphrased)

Air is being pumped into a spherical weather balloon. At time t, its volume is $V(t)$ and its radius is $r(t)$.

(a) Explain what the derivatives $\frac{dV}{dr}$ and $\frac{dV}{dt}$ represent.

(b) Express $\frac{dV}{dt}$ in terms of $\frac{dr}{dt}$. mymathclasse...

Part (a): Interpret the derivatives

1. What does $\frac{dV}{dr}$ represent?

It represents how fast the balloon's volume changes **with respect to its radius**.

For example, if the radius increases slightly, $\frac{dV}{dr}$ tells us approximately how much the volume increases per unit increase in radius.

Units:

$$\frac{\text{cm}^3}{\text{cm}} = \text{cm}^2$$

2. What does $\frac{dV}{dt}$ represent?

It represents how fast the balloon's volume changes **with respect to time**.

In other words, it is the rate at which the balloon is being inflated.

Units might be:

$$\text{cm}^3/\text{s}$$

Part (b): Express $\frac{dV}{dt}$ in terms of $\frac{dr}{dt}$

Step 1: Write the volume formula for a sphere.

$$\boxed{V = \frac{4}{3}\pi r^3}$$

Step 2: Differentiate with respect to time t.

Since the radius r changes with time, use the Chain Rule:

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \frac{dr}{dt}$$

Simplify:

$$\boxed{\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}}$$

Final answers

(a)

$$\frac{dV}{dr} : \text{ Volume change per unit radius}$$

$$\frac{dV}{dt} : \text{ Volume change per unit time}$$

(b)

$$\boxed{\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}}$$

Key concept for Jerry

The main lesson is understanding how the **Chain Rule connects two different rates**.

Since:

$$V = V(r), \qquad r = r(t)$$

Volume depends on radius, and radius depends on time.

Therefore:

$$\boxed{\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}}$$

In this problem:

$$\frac{dV}{dr} = 4\pi r^2$$

So:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

An interesting observation: $4\pi r^2$ is also the formula for a sphere's surface area.

This means:

$$\boxed{\text{Volume growth rate} = \text{Surface area} \times \text{Radius growth rate}}$$

Jerry should understand this relationship rather than memorize the entire related-rates formula. It is a direct application of the Chain Rule and can be derived whenever needed.

You said:

problem 81 statement and solution

ChatGPT said:

Homework #35 — Page 199, Problem 81

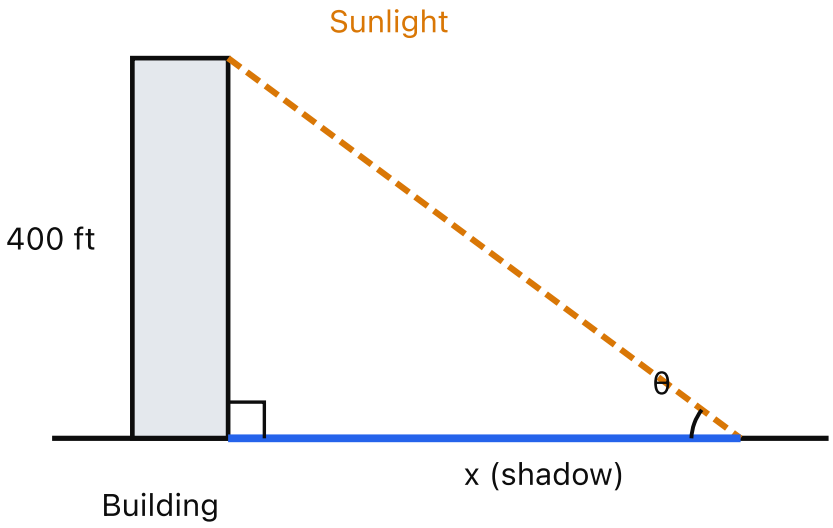
Textbook: James Stewart, *Calculus*, 8th Edition
Topic: Related Rates — Trigonometry and Chain Rule

Problem statement (paraphrased)

The angle of elevation of the sun is decreasing at a rate of **0.25 radians per hour**.

How fast is the shadow cast by a **400-foot-tall building** increasing when the angle of elevation of the sun is $\pi/6$? GradeSaver

Step 1: Draw a diagram



Let:

- x = length of the building's shadow (feet)
- θ = angle of elevation of the sun (radians)

Given:

$$\frac{d\theta}{dt} = -0.25 \text{ rad/h}$$

The rate is negative because the sun's angle is decreasing.

Find:

$$\boxed{\frac{dx}{dt} \text{ when } \theta = \frac{\pi}{6}}$$

Step 2: Set up a trigonometric relationship

Using the right triangle:

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\boxed{\tan\theta = \frac{400}{x}}$$

Solve for x:

$$x = \frac{400}{\tan\theta}$$

Since $\cot\theta = 1/\tan\theta$:

$$\boxed{x = 400\cot\theta}$$

Step 3: Differentiate with respect to time

Recall:

$$\frac{d}{d\theta}\cot\theta = -\csc^2\theta$$

Using the Chain Rule:

$$\boxed{\frac{dx}{dt} = -400\csc^2\theta\frac{d\theta}{dt}}$$

Step 4: Substitute the values

At:

$$\theta = \frac{\pi}{6}$$

We know:

$$\sin\frac{\pi}{6}=\frac{1}{2}$$

Therefore:

$$\csc\frac{\pi}{6}=2$$

$$\csc^2\frac{\pi}{6}=4$$

Substitute:

$$\frac{dx}{dt}=-400(4)(-0.25)$$

$$\frac{dx}{dt}=400$$

Final answer

$$\boxed{\frac{dx}{dt}=400\ \text{ft/h}}$$

The building's shadow is increasing in length at **400 feet per hour** when the sun's angle of elevation is $\pi/6$.

Key concepts for Jerry

1. The hardest step is setting up the trigonometric relationship.

$$\tan\theta=\frac{400}{x}$$

Once Jerry recognizes this relationship, the rest is mainly applying the Chain Rule.

2. Remember the negative derivative of cotangent.

$$\boxed{\frac{d}{dx}\cot x = -\csc^2 x}$$

Jerry can derive it from:

$$\cot x = \frac{\cos x}{\sin x}$$

Using the Quotient Rule:

$$\frac{d}{dx}\cot x = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{1}{\sin^2 x} = -\csc^2 x$$

3. Check the signs.

Because the sun is getting lower, its angle decreases:

$$\frac{d\theta}{dt} < 0$$

But the shadow becomes longer:

$$\frac{dx}{dt} > 0$$

The answer is positive, which makes physical sense.

AP Calculus exam tip: Don't substitute $\theta = \pi/6$ before differentiating. Keep θ as a changing variable until the related-rates equation has been established.

You said:

problem 80 is "A waterskier...", check again and solve it

Worked for 32s >

ChatGPT said:

Homework #35 — Page 199, Problem 80

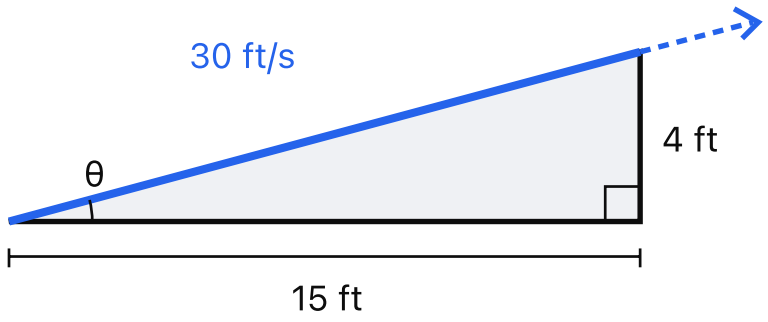
You're correct. I previously identified the wrong problem. I found the waterskier exercise in Stewart's textbook. web.lums.edu....

Problem statement

A waterskier travels up a ramp at a speed of **30 ft/s**. The ramp has a horizontal length of **15 ft** and a vertical height of **4 ft**.

How fast is the waterskier rising as she leaves the ramp?

Step 1: Understand the geometry



Schematic of the ramp, not to scale.

The waterskier's speed along the ramp is 30 ft/s. We need the **vertical component of that velocity**.

Step 2: Find the ramp's length

Using the Pythagorean theorem:

$$L = \sqrt{15^2 + 4^2}$$

$$\boxed{L = \sqrt{241}\ \text{ft}}$$

Step 3: Find the vertical rate

Let s represent distance traveled along the ramp, and y represent vertical height.

From similar triangles:

$$\frac{y}{s} = \frac{4}{\sqrt{241}}$$

Therefore:

$$y = \frac{4}{\sqrt{241}}s$$

Differentiate with respect to time:

$$\boxed{\frac{dy}{dt} = \frac{4}{\sqrt{241}}\frac{ds}{dt}}$$

We know:

$$\frac{ds}{dt} = 30\ \text{ft/s}$$

Substitute:

$$\frac{dy}{dt} = \frac{4}{\sqrt{241}}(30)$$

Final answer

$$\boxed{\frac{dy}{dt} = \frac{120}{\sqrt{241}} \approx 7.73 \text{ ft/s}}$$

The waterskier is rising at approximately **7.73 feet per second** as she leaves the ramp.

Key concept for Jerry

This is a velocity-components problem that can be solved with trigonometry.

Since:

$$\sin\theta = \frac{4}{\sqrt{241}}$$

The vertical component of velocity is:

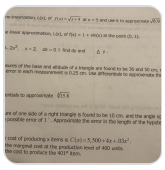
$$\boxed{v_y = v\sin\theta}$$

$$v_y = 30\left(\frac{4}{\sqrt{241}}\right)$$

$$\boxed{v_y \approx 7.73 \text{ ft/s}}$$

The important distinction is that **30 ft/s is the speed along the inclined ramp, not the vertical speed**.

You said:



here are the problem 4-10 in RR worksheet:

ChatGPT said:

I can read the RR worksheet problems #4–10 from the photo. For Jerry's **Homework #35**, the assigned worksheet problems are **#7, #8, and #9**.

Here are their problem statements and step-by-step solutions.

RR Worksheet — Problem 7

Problem statement

The measures of the base and altitude of a triangle are found to be 36 and 50 cm, respectively. The possible error in each measurement is 0.25 cm. Use differentials to approximate the possible error in area.

Step 1: Write the area formula

$$A = \frac{1}{2}bh$$

Given:

$$b = 36, \quad h = 50$$

$$|db| = 0.25, \quad |dh| = 0.25$$

Step 2: Find the total differential

Because both b and h can change, use the product rule:

$$\boxed{dA = \frac{1}{2}(h\,db + b\,dh)}$$

Step 3: Substitute

For the maximum approximate error, take both errors in the same direction:

$$|dA| \approx \frac{1}{2}[50(0.25) + 36(0.25)]$$

$$= \frac{1}{2}(12.5 + 9)$$

Final answer

$$\boxed{|dA| \approx 10.75 \text{ cm}^2}$$

Key concept: When a quantity depends on two measurements, both measurement errors contribute to its total differential.

RR Worksheet — Problem 8

Problem statement

Use differentials to approximate:

$$\boxed{\sqrt[4]{15.8}}$$

Step 1: Choose a function

$$f(x) = x^{1/4}$$

Choose $a = 16$ because:

$$\sqrt[4]{16} = 2$$

Therefore:

$$dx = 15.8 - 16 = -0.2$$

Step 2: Differentiate

$$f'(x) = \frac{1}{4}x^{-3/4}$$

At $x = 16$:

$$f'(16) = \frac{1}{4(16^{3/4})}$$

Since $16^{3/4} = 8$:

$$\boxed{f'(16) = \frac{1}{32}}$$

Step 3: Apply differentials

$$dy = f'(x)\,dx$$

$$dy = \frac{1}{32}(-0.2)$$

$$dy = -0.00625$$

Thus:

$$f(15.8) \approx f(16) + dy$$

$$= 2 - 0.00625$$

Final answer

$$\boxed{\sqrt[4]{15.8} \approx 1.99375}$$

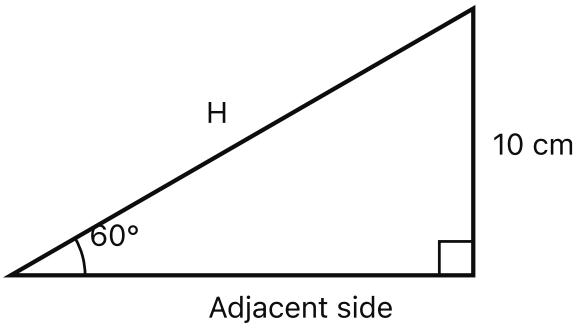
Key concept: Choose a nearby perfect fourth power, and use $f(a) + dy$ to approximate the desired value.

RR Worksheet — Problem 9

Problem statement

The measure of one side of a right triangle is found to be 10 cm, and the angle opposite that side is 60°, with a possible error of 1°. Approximate the error in the length of the hypotenuse.

Step 1: Establish the geometry



Let H be the hypotenuse and θ the angle opposite the 10-cm side.

Using sine:

$$\sin\theta = \frac{10}{H}$$

Solve for H:

$$\boxed{H = \frac{10}{\sin\theta} = 10\csc\theta}$$

Step 2: Differentiate

$$\frac{dH}{d\theta} = -10\csc\theta\cot\theta$$

Therefore:

$$\boxed{dH = -10\csc\theta\cot\theta\,d\theta}$$

Step 3: Convert the angle error into radians

The possible error is 1°.

$$|d\theta| = \frac{\pi}{180}$$

At $\theta = 60^\circ = \pi/3$:

$$\csc 60^\circ = \frac{2}{\sqrt{3}}$$

$$\cot 60^\circ = \frac{1}{\sqrt{3}}$$

Step 4: Calculate the approximate error

$$|dH| = 10\left(\frac{2}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right)\left(\frac{\pi}{180}\right)$$

$$= \frac{20}{3} \cdot \frac{\pi}{180}$$

$$= \frac{\pi}{27}$$

Final answer

$$\boxed{|dH| \approx 0.1164 \text{ cm}}$$

The possible error in the hypotenuse is approximately **0.116 cm**.

Key concept: Trigonometric differentials require angle changes to be expressed in radians. The negative derivative indicates that the hypotenuse becomes shorter as the opposite angle increases, assuming the 10-cm side stays fixed.

Summary for Jerry

Problem	Main technique	Final answer
7	Total differential, product rule	10.75 cm^2
8	Linear approximation	1.99375
9	Trigonometric differential	0.1164 cm

Problem 9 is the most challenging of these three because Jerry needs to establish a trigonometric relationship, differentiate it, and convert degrees to radians.

All three problems reinforce an important concept:

$$\boxed{\Delta y \approx dy}$$

The differential gives an approximation to the actual change or measurement error, not necessarily the exact change.

You said:

recap problem 4-6 and solve them

ChatGPT said:

RR Worksheet — Problems 4–6

These three problems review **linearization, linear approximation, and differentials**, which are important concepts in Jerry's AP Calculus AB Homework #35.

Problem 4 — Linearization and approximation

Problem statement

Find the linearization, $L(x)$, of

$$f(x) = \sqrt{x+4}$$

at $x = 5$ and use it to approximate:

(a) $\sqrt{8.9}$

(b) $\sqrt{9.2}$

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x-a)}$$

Here:

$$a = 5$$

Step 2: Calculate $f(a)$

$$f(5) = \sqrt{5+4} = \sqrt{9} = 3$$

Step 3: Find the derivative

$$f(x) = (x+4)^{1/2}$$

Using the Chain Rule:

$$f'(x) = \frac{1}{2}(x+4)^{-1/2}$$

$$\boxed{f'(x) = \frac{1}{2\sqrt{x+4}}}$$

At $x = 5$:

$$f'(5) = \frac{1}{2\sqrt{9}} = \frac{1}{6}$$

Step 4: Write the linearization

$$L(x) = 3 + \frac{1}{6}(x-5)$$

Therefore:

$$\boxed{L(x) = 3 + \frac{x-5}{6}}$$

Part (a): Approximate $\sqrt{8.9}$

We need to find x such that:

$$x + 4 = 8.9$$

$$x = 4.9$$

Substitute into $L(x)$:

$$L(4.9) = 3 + \frac{4.9 - 5}{6}$$

$$= 3 - \frac{0.1}{6}$$

$$\boxed{\sqrt{8.9} \approx 2.98333}$$

Part (b): Approximate $\sqrt{9.2}$

Set:

$$x + 4 = 9.2$$

$$x = 5.2$$

Substitute:

$$L(5.2) = 3 + \frac{5.2 - 5}{6}$$

$$= 3 + \frac{0.2}{6}$$

$$\boxed{\sqrt{9.2} \approx 3.03333}$$

Final answers

$$\boxed{L(x) = 3 + \frac{x-5}{6}}$$

$$\boxed{\sqrt{8.9} \approx 2.98333}$$

$$\boxed{\sqrt{9.2} \approx 3.03333}$$

Key concept for Jerry: The input to $f(x)$ is not the same as the number under the square root. Because $f(x) = \sqrt{x+4}$, we must subtract 4 from 8.9 and 9.2 before substituting into $L(x)$.

Problem 5 — Linear approximation of a trigonometric function

Problem statement

Find the linear approximation, $L(x)$, of

$$f(x) = 1 + \sin x$$

at the point $(0, 1)$.

Step 1: Identify the base point

The point $(0, 1)$ tells us:

$$a = 0, \qquad f(0) = 1$$

Check:

$$f(0) = 1 + \sin 0 = 1$$

Step 2: Find the derivative

$$f(x) = 1 + \sin x$$

Differentiate:

$$\boxed{f'(x) = \cos x}$$

Evaluate at $x = 0$:

$$f'(0) = \cos 0 = 1$$

Step 3: Apply the linearization formula

$$L(x) = f(a) + f'(a)(x - a)$$

Substitute:

$$L(x) = 1 + 1(x - 0)$$

Final answer

$$\boxed{L(x) = 1 + x}$$

This means that near $x = 0$:

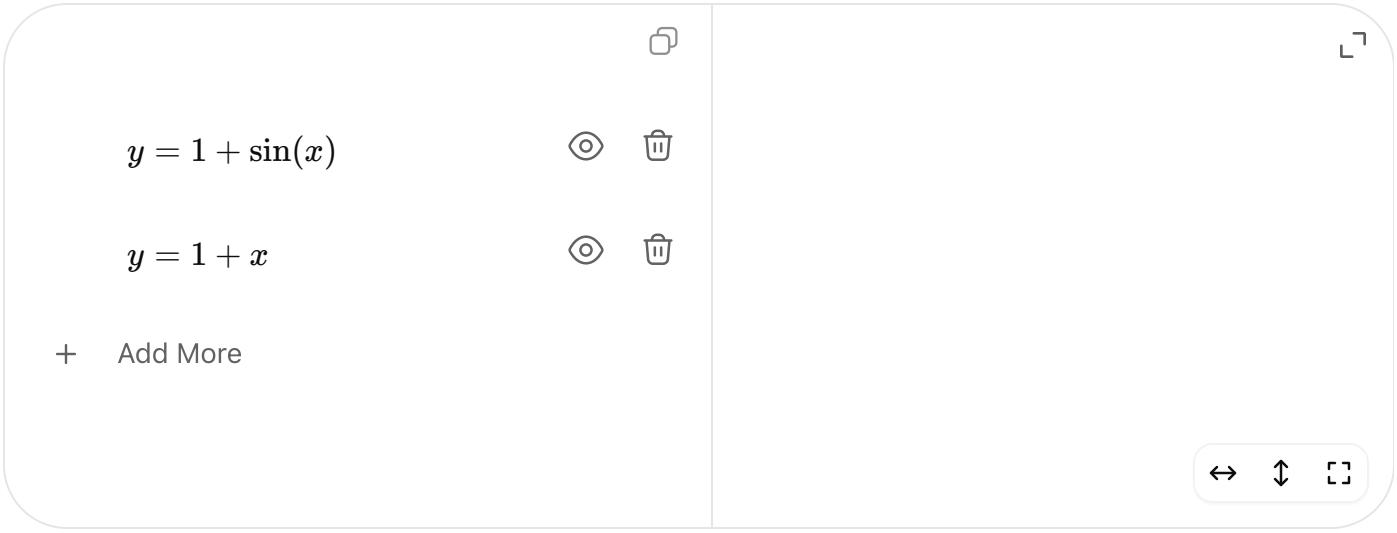
$$\boxed{1 + \sin x \approx 1 + x}$$

Or equivalently:

$$\boxed{\sin x \approx x}$$

This last approximation is valid when x is close to zero and measured in radians.

Visual explanation



Give feedback

The straight line $y = 1 + x$ is tangent to $y = 1 + \sin x$ at $(0, 1)$. The two functions are very close near $x = 0$, but differ more as x moves farther away.

Key concept for Jerry: Linearization is simply the equation of the tangent line, written as a function:

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Problem 6 — Differentials versus actual change

Problem statement

If

$$y = 1 - 2x^2, \qquad x = 2, \qquad dx = 0.1$$

find:

(a) dy

(b) Δy

Part (a): Find dy

Step 1: Differentiate the function.

$$y = 1 - 2x^2$$

$$y' = -4x$$

Step 2: Use the differential formula.

$$\boxed{dy = y'\,dx}$$

Therefore:

$$dy = -4x\,dx$$

Substitute $x = 2$ and $dx = 0.1$:

$$dy = -4(2)(0.1)$$

$$\boxed{dy = -0.8}$$

Part (b): Find Δy

Remember:

$$\boxed{\Delta y = f(x + \Delta x) - f(x)}$$

Since:

$$\Delta x = dx = 0.1$$

and $x = 2$:

$$\Delta y = f(2.1) - f(2)$$

Step 1: Calculate $f(2)$.

$$f(2) = 1 - 2(2)^2$$

$$= 1 - 8 = -7$$

Step 2: Calculate $f(2.1)$.

$$f(2.1) = 1 - 2(2.1)^2$$

$$= 1 - 2(4.41)$$

$$= 1 - 8.82 = -7.82$$

Step 3: Calculate the actual change.

$$\Delta y = -7.82 - (-7)$$

$$\boxed{\Delta y = -0.82}$$

Final answers

Differential (approximation)

−0.8

$dy = -0.8$

Actual change

−0.82

$\Delta y = -0.82$

Why are dy and Δy different?

- dy estimates the change using the tangent line.
- Δy measures the actual change using the original function.

Their difference here is:

$$|\Delta y - dy| = |-0.82 - (-0.8)|$$

$$\boxed{0.02}$$

The difference exists because $y = 1 - 2x^2$ is a curved function rather than a straight line.

Summary: Problems 4–6

Problem	Key concept	Final answer
4	Linearization of square root	$L(x) = 3 + \frac{x-5}{6}$; $2.98333,\ 3.03333$
5	Tangent-line approximation	$L(x) = 1 + x$
6	Differential vs. actual change	$dy = -0.8,\ \Delta y = -0.82$

The most important distinction Jerry should understand is:

$$\boxed{dy = f'(x)\,dx}$$

versus

$$\boxed{\Delta y = f(x + \Delta x) - f(x)}$$

For AP Calculus tests: dy is the tangent-line approximation, while Δy is the exact change. They are generally close when dx is small, but they are not necessarily equal.